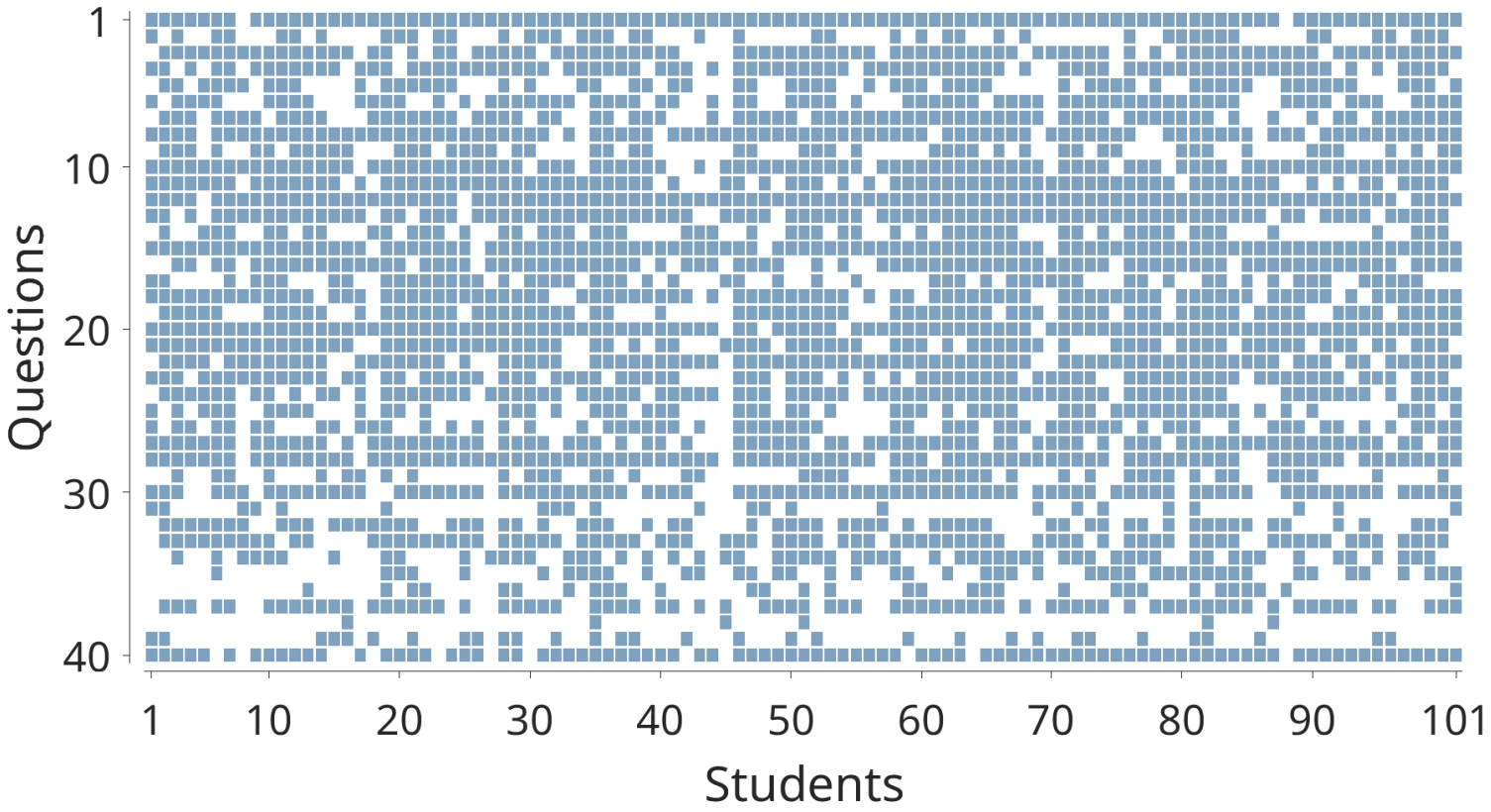
The Rasch Model

Michael D. Lee

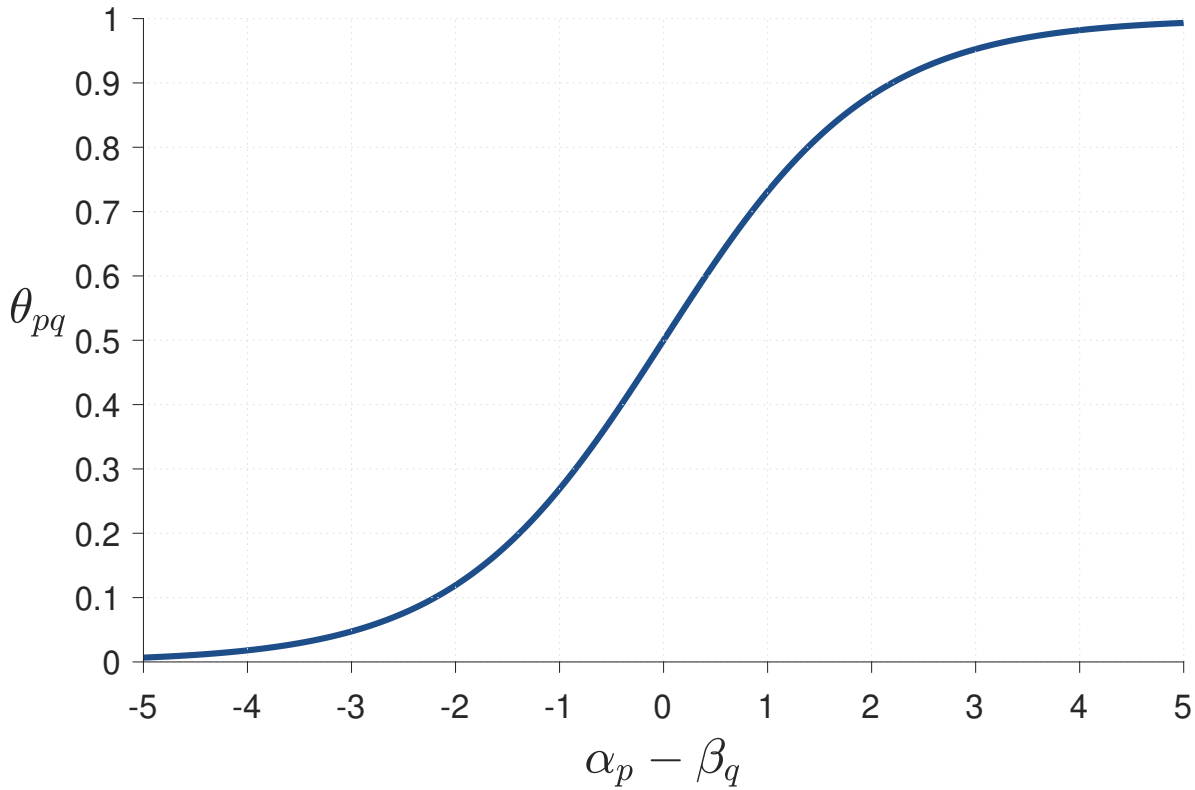
The Rasch Model

- The Rasch model is the simplest form of a more general class of psychometric models known as item response theory models (Wilson, De Boeck, & Carstensen, 2008)
  - motivated by the problem of measuring people's ability on the basis of their performance in evaluative tests



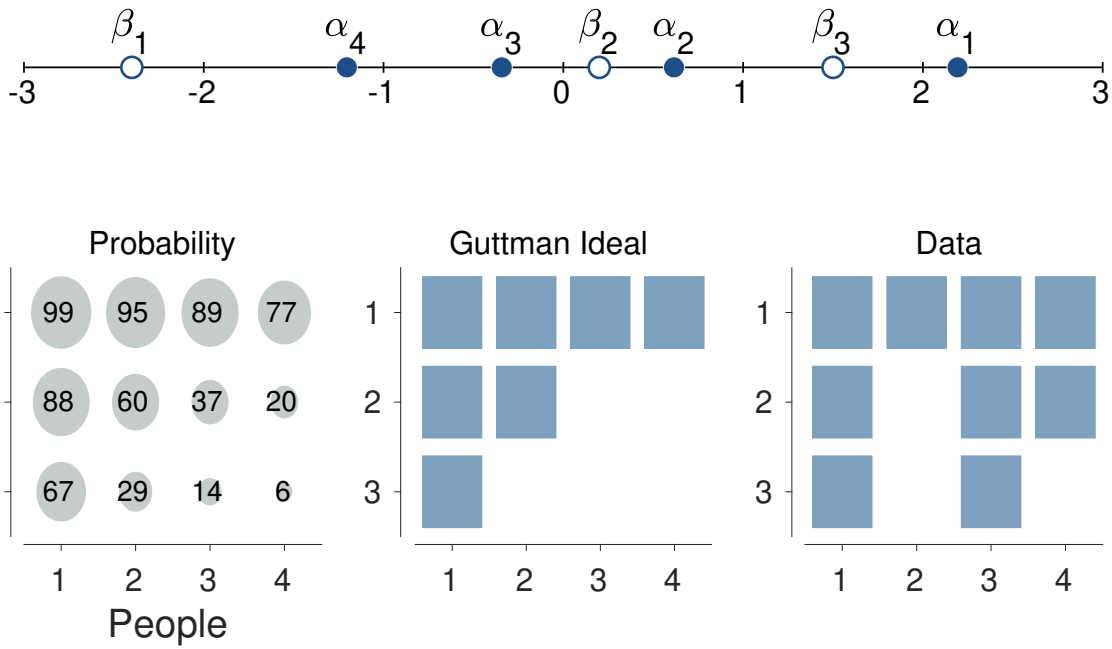
Ability and Difficulty

- The Rasch model assumes that
  - each person has a latent ability α_p
  - each question has a latent difficulty β_q
  - the difference between ability and difficulty $\alpha_p = \beta_q$ is called the dominance and determines the probability θ_{pq} that they will answer the question correctly



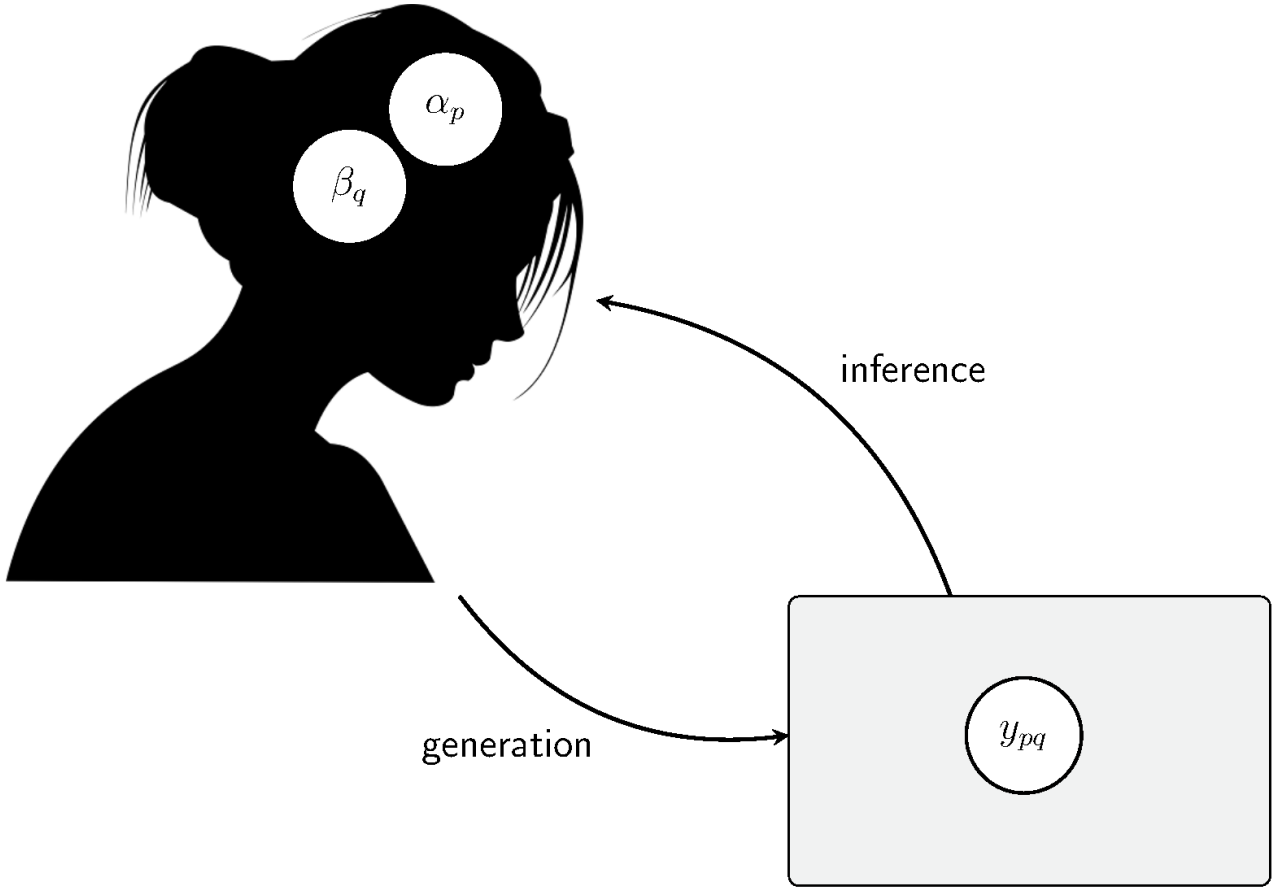
Guttman Ideal

- The Rasch model's assumption that abilities and difficulties lie on the same scale constrains the predictions it makes about patterns of responses
  - e.g., if person 1 has greater ability than person 2 they have a greater probability of answering all items correctly



Rasch Model Structure

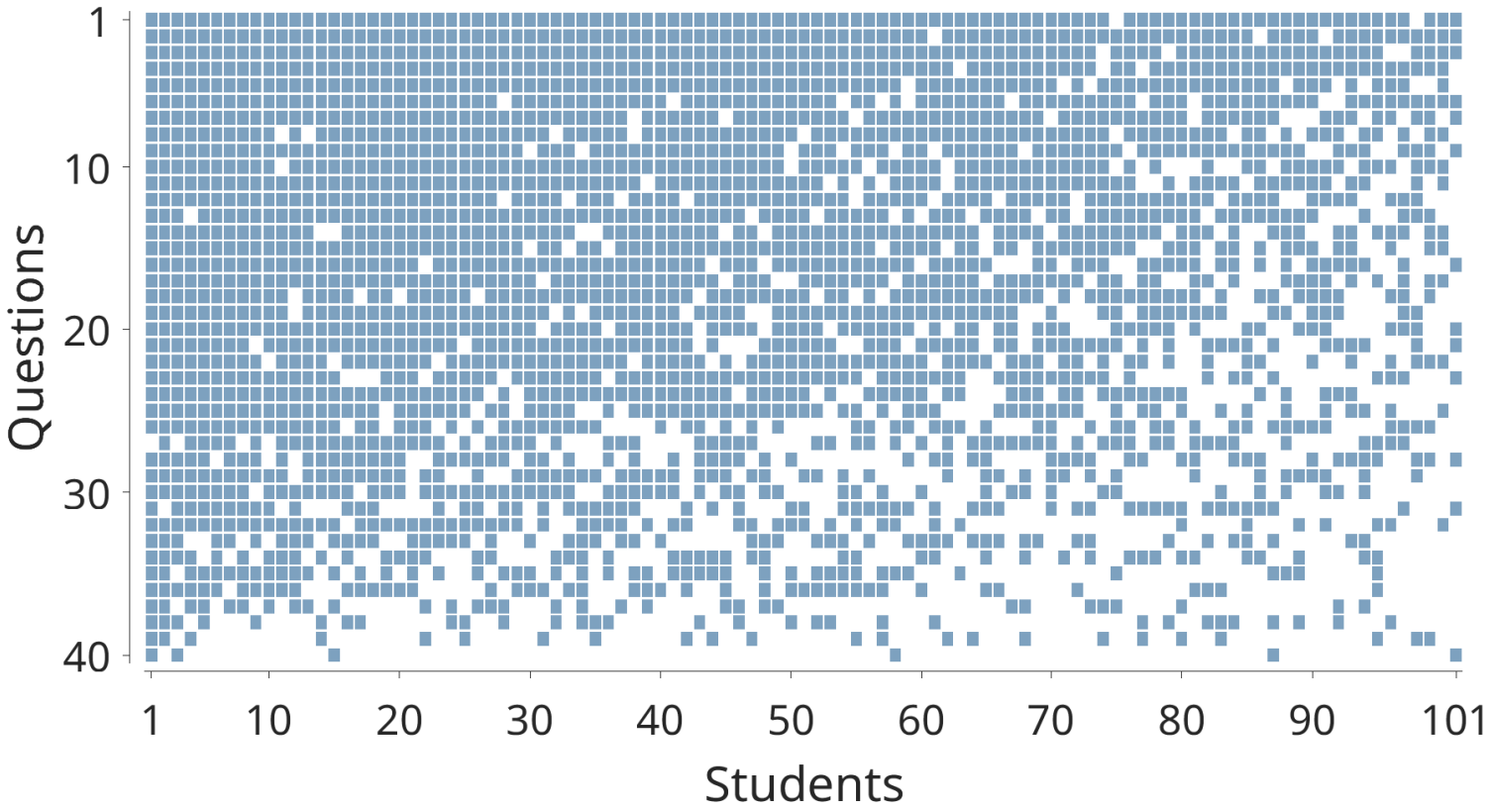
- The Rasch model has the structure
  - observed binary decisions y_{pq} for person p answering question q
  - person ability α_p and question difficulty β_q parameters



Class Exam

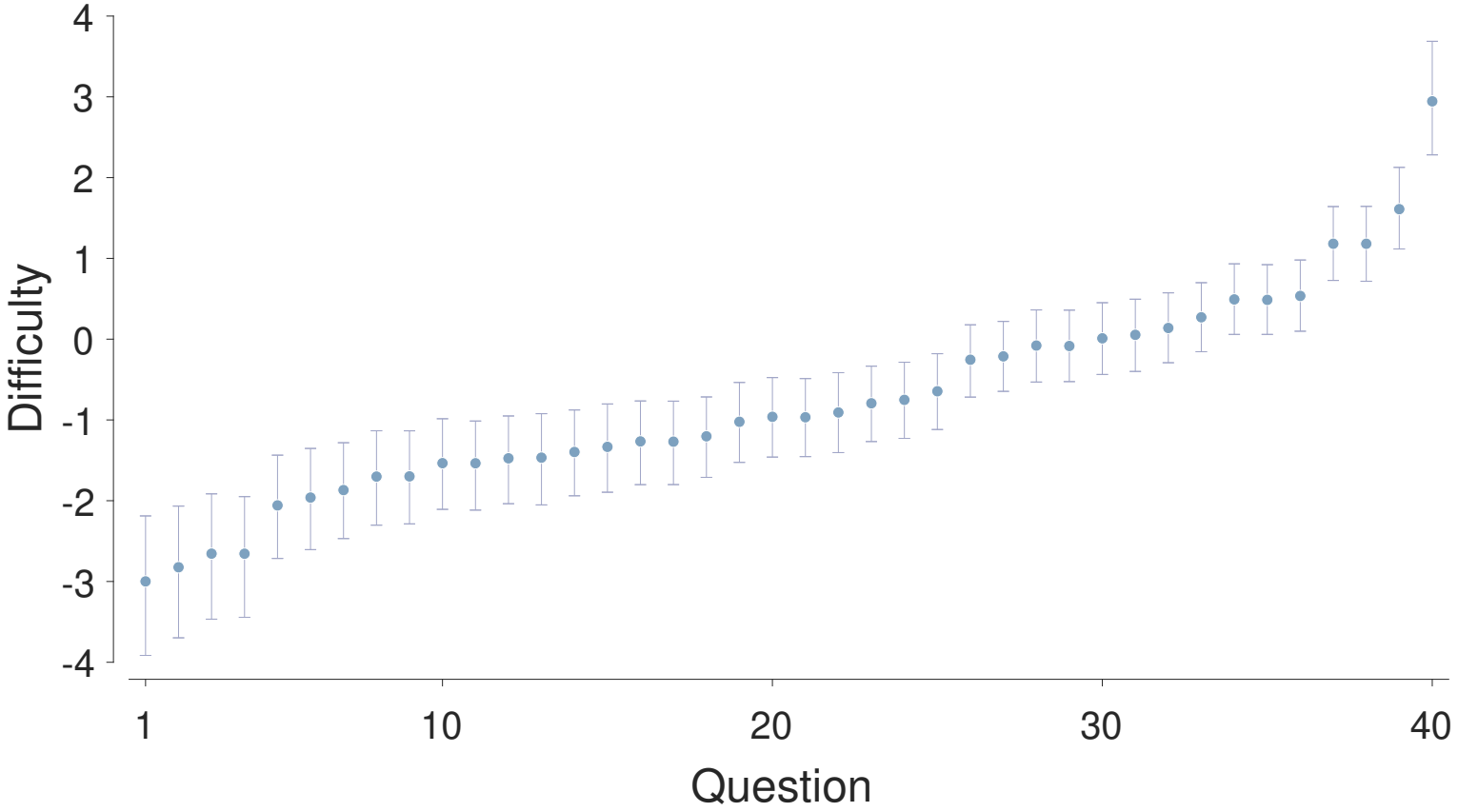
Class Exam Data

- The figure shows the same class data as earlier
  - reorganized in terms of correctness for both people and questions
  - provides a visual check of the Guttman Ideal assumptions



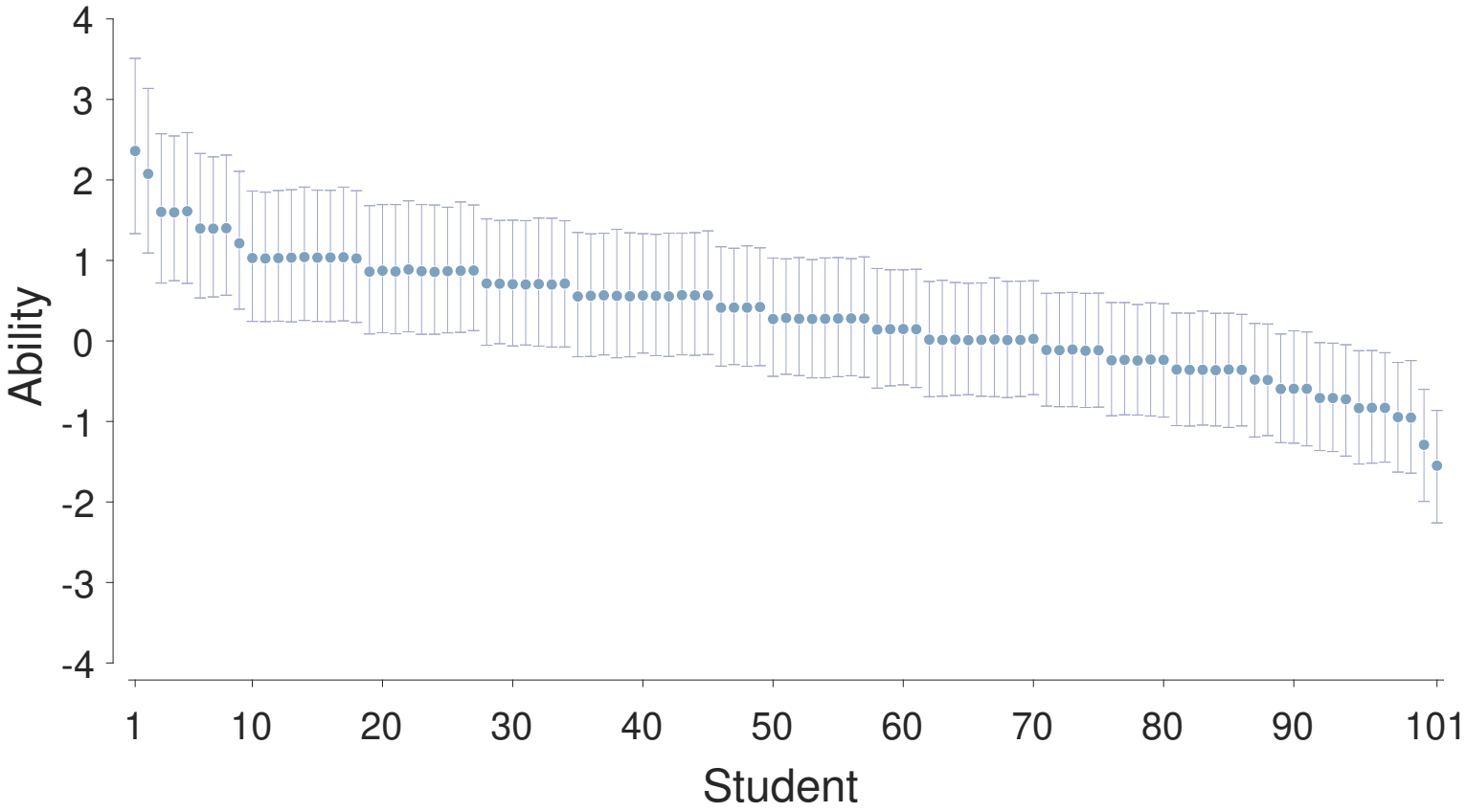
Inferred Question Difficulties

- The inferred difficulties follow the number of times questions were answered correctly
  - but provide a continuous measurement that is sensitive to which people got the question right and wrong



Inferred Person Abilities

- The inferred abilities follow the number of questions each person answered correctly
  - but provide a continuous measurement that is sensitive to which questions they got right and wrong



Cricket Santa

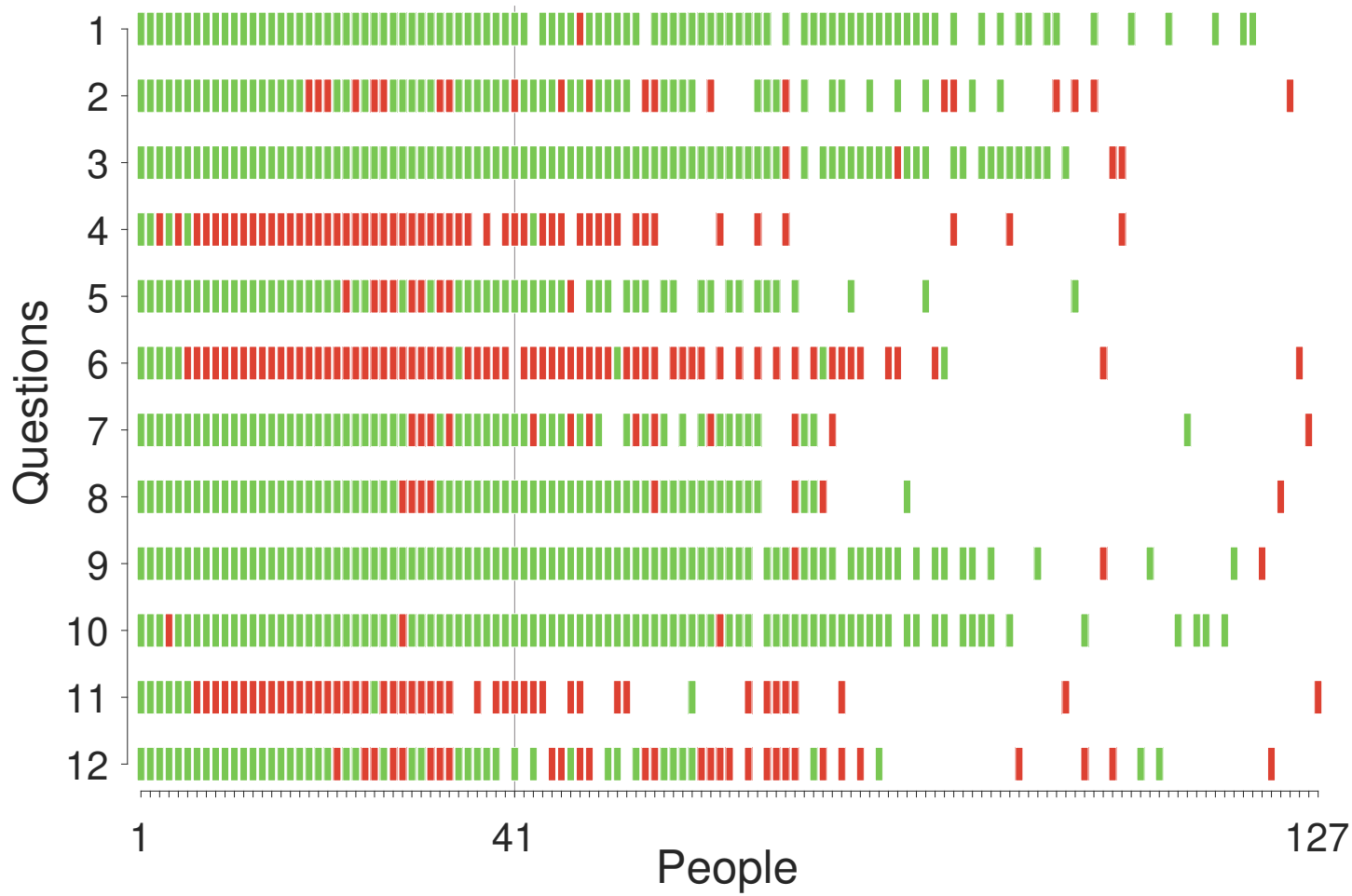
A Cricket Quiz

- An online quiz asked people to identify the 12 cricket bowlers, based on their bowling actions as depicted in the figure shown



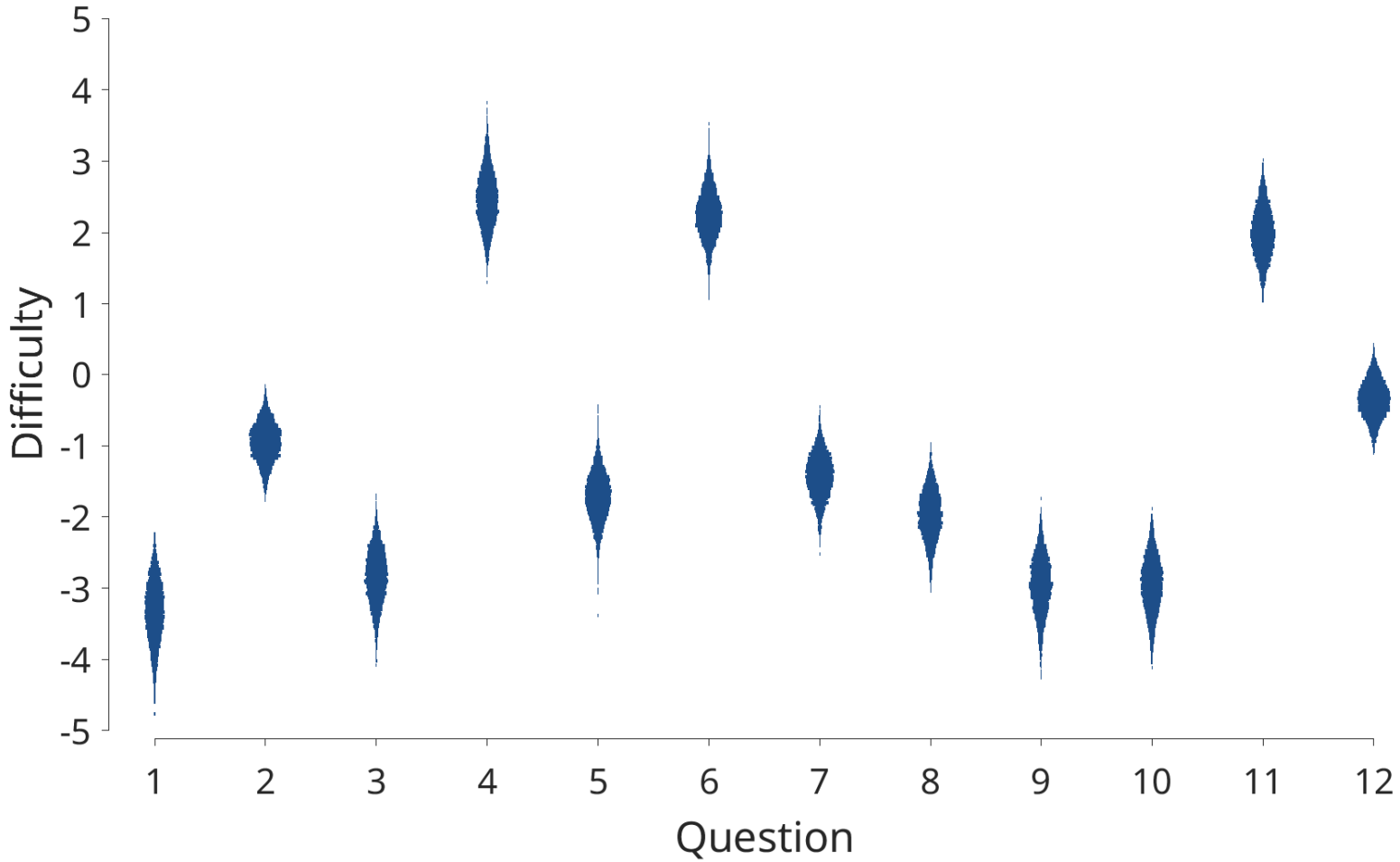
Data

- Data were collected from twitter responses, collating a data set of 127 people who provided a total of 925 out of 1524 possible answers
  - green markers indicate a person answered a questions correctly, and a red marker indicates an incorrect answer



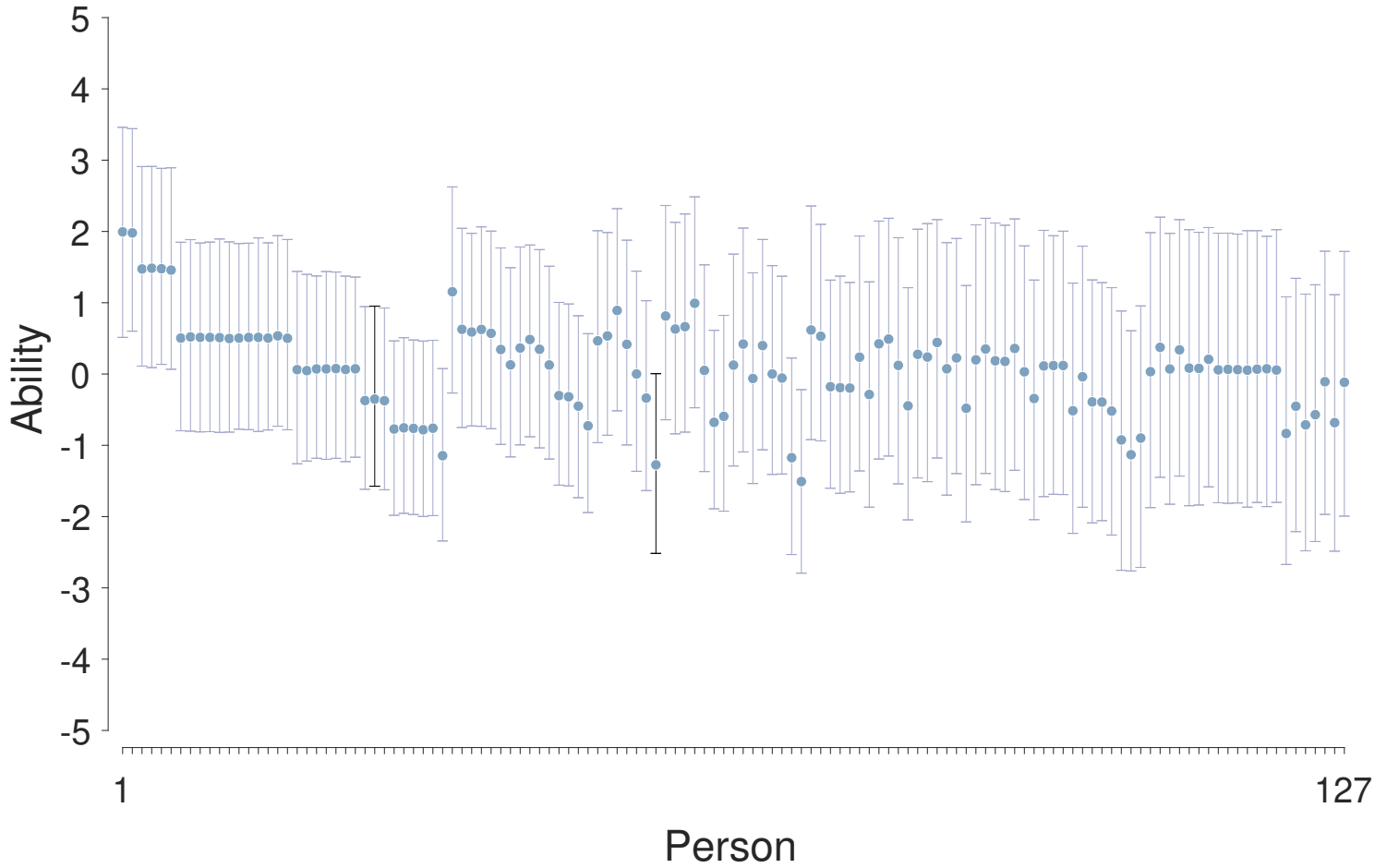
Inferred Question Difficulties

- The infererred difficulties match intuitions
  - questions 4, 6, and 11 are difficult
  - questions 1, 3, 9, and 10 are easy



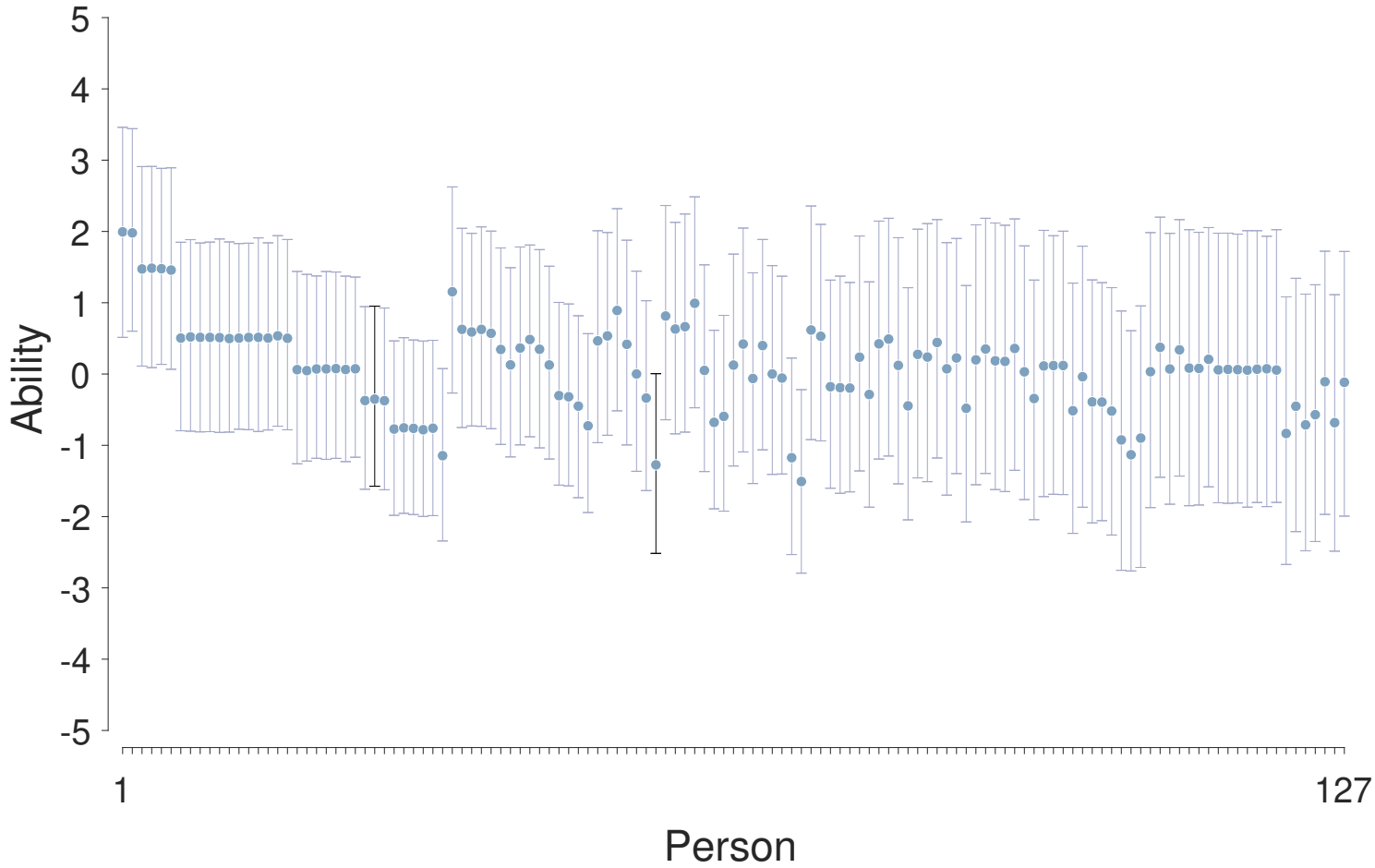
Inferred Person Abilities

- The inferred abilities depend on whether answers were correct or not, but also on which questions were attempted
  - people from 122 to 127 answered one question incorrectly, but have differences in inferred ability depending on which question it was



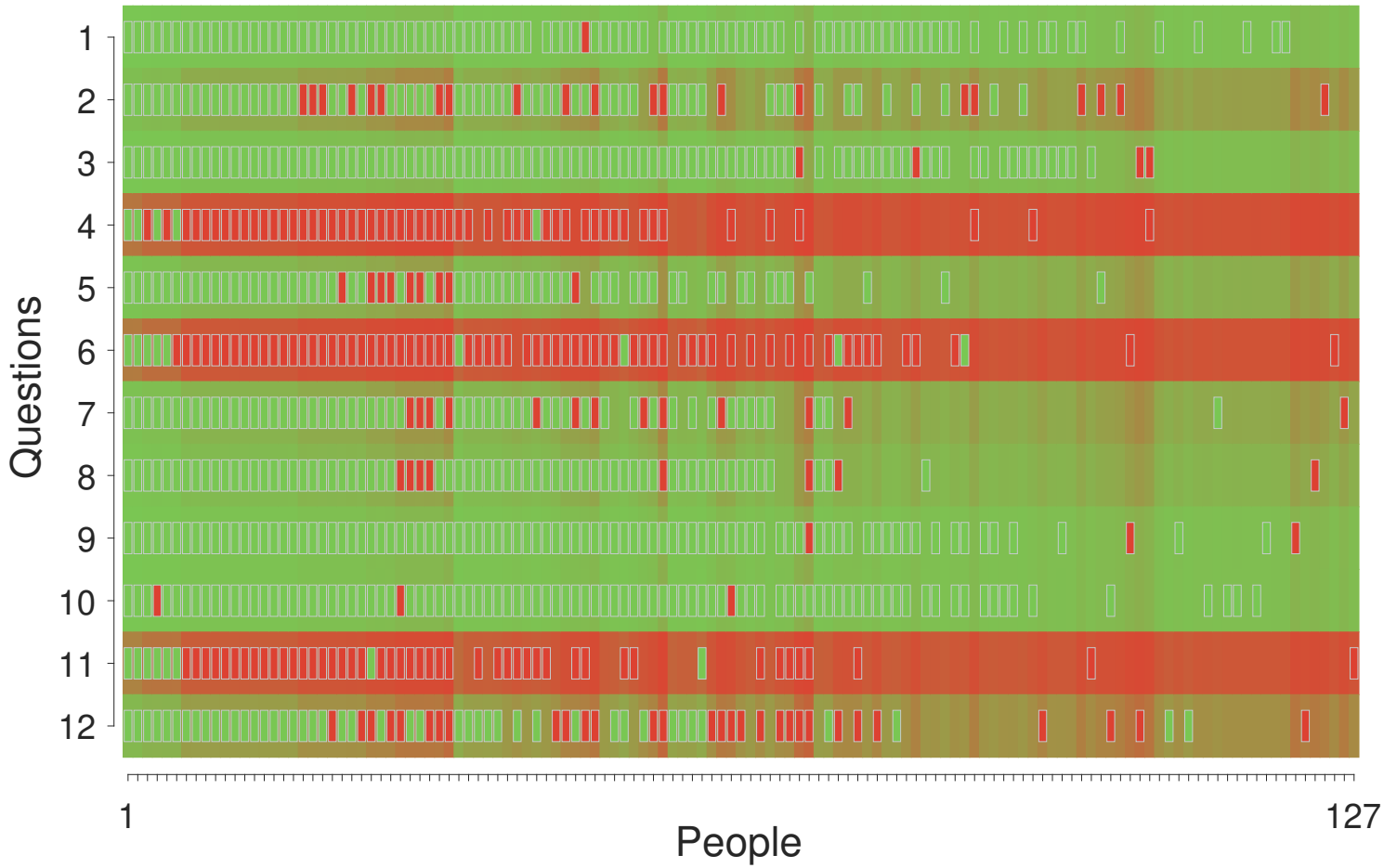
Inferred Person Abilities

- The inferred abilities depend on whether answers were correct or not, but also on which questions were attempted
  - person 41 and 42 both attempted 11 questions and got 3 wrong, but have interpretable differences in inferred ability



Posterior Predictive Analysis

- The posterior predictive distribution
  - tests descriptive adequacy on the 925 answers actually provided
  - makes predictions about the 599 person-question combinations that were not answered



References i

Wilson, M., De Boeck, P., & Carstensen, C. H. (2008). Explanatory item response models: A brief introduction. *Assessment of competencies in educational contexts*, 91–120.